Card Fraud Triage via Gaussian Mixture Density Scoring and Unsupervised Anomaly Detection

Phase 1 Research Monograph: Methodological Foundations, Validation Protocol, and Out-of-Time Test Benchmarking

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***Abstract*—**Financial card fraud detection in production banking environments faces severe class imbalance (~0.17%), delayed verification ground truth (often 30–90 days for customer disputes), and strict constraints on human investigation bandwidth. While supervised classifiers degrade when fraud topologies evolve, unsupervised density estimation provides an operational triage filter by modelling purely legitimate consumer behavior. In this Phase 1 study, we investigate whether a Gaussian Mixture Model (GMM) trained exclusively on normal transactions can effectively rank fraudulent swipes for manual triage under realistic operational constraints. Using the benchmark ULB European cardholder dataset (283,726 deduplicated transactions across 48 hours), we enforce strict temporal integrity via a 60/20/20 chronological split. Feature distributions reject univariate normality ($p < 10^{-100}$), confirming multimodality. Training exclusively on 169,893 normal transactions from the first 60% of time, we execute an extensive grid search across component counts (K = 1 to 8) and random initialisations. The champion configuration (K = 3, Full Covariance, 29 features) achieves an Average Precision (AP) of 0.612 (95% bootstrap CI [0.485, 0.743]) on the completely unseen test period, outperforming the random-guess baseline (0.0013) by 470×. At an operational false-positive rate budget fixed at ≤ 0.5% on validation ($\tau = 94.68$), the model intercepts 56 of 74 test frauds (75.7% Recall) with only 158 false alarms (0.28% FPR). In human review triage scenarios, the top 50 flagged transactions achieve an alert precision of 84.0% (42 confirmed frauds). Our empirical findings demonstrate that full-covariance mixture models capture subtle multivariate cross-correlations that univariate feature selection discards, establishing an unsupervised baseline for ongoing Phase 2 autoencoder research.

***Index Terms*—**Fraud detection, unsupervised anomaly detection, Gaussian mixture models, expectation-maximization, class imbalance, precision-recall curve, operational triage.

I. INTRODUCTION

Credit card fraud causes tens of billions of dollars in annual global losses, posing an existential challenge to financial payment processors and consumer banking infrastructure [1]. Designing automated fraud defense mechanisms involves three fundamental characteristics of transactional data:

1. **Extreme Class Imbalance:** Fraudulent activities constitute a minuscule fraction of daily volume (typically under 0.2%). Standard accuracy is uninformative, as a static null classifier predicting all transactions as legitimate achieves >99.8% accuracy while failing completely.
2. **Delayed Verification Ground Truth:** In production environments, fraud labels are not immediately available. Fraud is typically reported weeks later via chargebacks, merchant reconciliations, or customer disputes [3]. Supervised classifiers that require contemporaneous labels suffer severe degradation under operational delay.
3. **Analyst Capacity Constraints:** Security operations centers have finite human review bandwidth. An algorithm generating thousands of low-confidence alerts overwhelms fraud investigation personnel, leading to alert fatigue and high operational costs.

To circumvent the latency of supervised label acquisition, unsupervised anomaly detection provides an operational alternative [4]. By training a density model strictly on abundant, verified historical legitimate purchases, the system establishes a probabilistic boundary of normal cardholder behavior. When incoming transactions exhibit low likelihood under the learned distribution, they receive high anomaly scores and are triaged into an analyst priority queue.

In this work (Phase 1 of Project 26), we develop and empirically validate an unsupervised fraud triage system utilizing Gaussian Mixture Models (GMM). We formulate anomaly detection through negative log-likelihood (NLL) density scoring. We evaluate our pipeline on the ULB credit card benchmark under a time-ordered chronological split, comparing full feature representations against dimensionality-reduced candidate sets.

II. RELATED WORK

Machine learning in credit card fraud detection has evolved across multiple paradigms. Dal Pozzolo et al. [1, 2] formalized the benchmark evaluation framework for card fraud, proving that random k-fold cross-validation introduces severe temporal lookahead leakage and inflates performance metrics. They established that chronological partitioning and human verification capacity metrics (Precision@k) are mandatory for realistic assessment.

Unsupervised anomaly detection has been surveyed comprehensively by Chandola et al. [4]. Classical methods include distance-based algorithms, linear projections, and parametric density models. Gaussian Mixture Models, optimized via Expectation-Maximization (EM) [5], offer analytical density evaluations and capture multimodal consumer spending habits through heterogeneous component centers and full covariance matrices.

Davis and Goadrich [6] established that under massive class skew, the Receiver Operating Characteristic (ROC) curve is structurally flattered by the overwhelming volume of True Negatives. They demonstrated that Precision-Recall (PR) curves and Average Precision (AP) provide an uncompromised assessment of minority-class retrieval.

Recent advancements explore deep autoencoding architectures [10] and deep autoencoding Gaussian mixture models (DAGMM) [7], which unify dimensionality reduction and density estimation. Our Phase 1 GMM baseline establishes the mathematical benchmark against which Phase 2 autoencoders will be contrasted.

III. DATASET AND TIME PARTITIONING

A. Dataset Properties

We utilize the public European cardholder benchmark dataset collected by the Machine Learning Group at Université Libre de Bruxelles (ULB) [2]. The dataset contains 284,807 transactions recorded over a continuous 48-hour period in September 2013, with 492 confirmed fraudulent events (0.173% prevalence).

The dataset consists of 30 numerical input features: `Time` (seconds from first transaction), `Amount` (in Euros), `V1`–`V28` (anonymized continuous numerical attributes derived from a proprietary Principal Component Analysis performed by the banking institution for data privacy), and `Class` (binary ground truth: 0 = Normal, 1 = Fraud).

B. Deduplication and Chronological Partitioning

Data auditing identified 1,081 exact duplicate rows (including 19 fraudulent records), which were purged to prevent artificial density inflation, leaving 283,726 clean rows.

TABLE I: CHRONOLOGICAL DATA PARTITIONING (60 / 20 / 20 SPLIT)

Split	Window	Rows	Frauds	Rate
Train	First 60% (0–28.8 h)	169,893	0 (342 rem.)	0.000%
Validation	Next 20% (28.8–38.4 h)	56,745	57	0.100%
Test	Last 20% (38.4–48.0 h)	56,746	74	0.130%

To prevent temporal data leakage, we enforce a strict 60/20/20 chronological partition based on `Time` (Table I):

1. **Training (First 60%):** 170,235 transactions. All 342 fraud records occurring in this window are removed. The remaining 169,893 purely legitimate transactions are used to fit the scaler and the GMM.
2. **Validation (Next 20%):** 56,745 transactions (57 frauds, 0.100% rate). Used to optimize hyperparameters (K, covariance constraints, random seeds) and tune the operational decision threshold.
3. **Test (Final 20%):** 56,746 transactions (74 frauds, 0.130% rate). Kept strictly sealed until model finalization for out-of-time evaluation.

IV. EXPLORATORY DATA ANALYSIS

All exploratory analysis and feature ranking are conducted exclusively on the training period without peeking at validation or test labels.

A. Diurnal Transaction Distributions

We derive transaction hour as: $Hour = \lfloor Time / 3600 \rfloor \bmod 24$. As illustrated in Fig. 1, normal transaction volume exhibits strong diurnal rhythm, dropping significantly between 01:00 and 06:00. Conversely, fraudulent swipes remain active during nocturnal hours, spiking between 02:00 and 04:00 where detection friction is minimal.

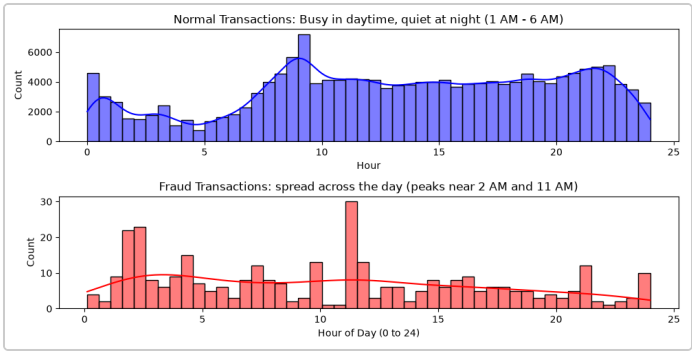


Fig. 1. Diurnal transaction volume distribution (Normal vs. Fraud) during the training period, demonstrating nocturnal fraud concentration.

B. Amount Log-Transformation

Raw purchase amounts span from €0.00 to €25,691.16 with a median of €22.00, yielding extreme positive skewness ($\gamma_1 = 15.77$). To accommodate the Gaussian structural assumptions of our mixture components, we apply a natural logarithmic transformation: $\tilde{Amount} = ln(1 + Amount)$.

Fig. 2 verifies that the transformation reduces skewness to $\gamma_1 = 0.16$, yielding an approximately symmetric bell-shaped distribution suitable for GMM covariance estimation.

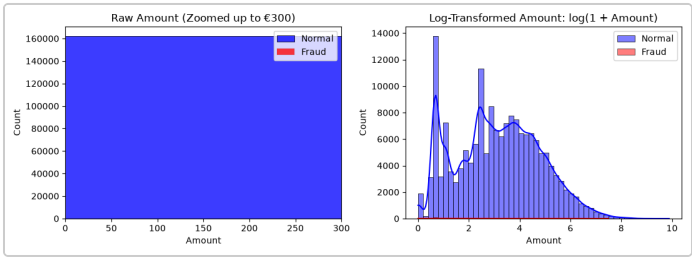


Fig. 2. Distribution of transaction Amount: Raw distribution exhibits extreme right-skew ($\gamma_1 = 15.77$), while log(1 + Amount) normalizes the profile ($\gamma_1 = 0.16$).

C. Feature Divergence Analysis (KS Test)

To identify the most discriminative features, we compute the two-sample Kolmogorov-Smirnov (KS) statistic [9] between normal and fraudulent training distributions: $D_KS = sup\ |F_normal(x) - F_fraud(x)|$.

The top 11 divergent features (V14, V10, V12, V11, V17, V4, V3, V16, V7, V2, V9) all exhibit D_KS > 0.32 (Fig. 3). These 11 features, combined with $\tilde{Amount}$, form our candidate 12-feature subset recorded in `configs/gmm_features.json`.

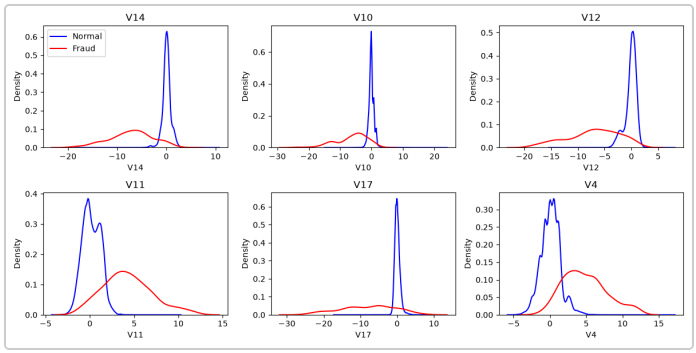


Fig. 3. Kernel density estimations for the top six separating features identified by the two-sample Kolmogorov-Smirnov test.

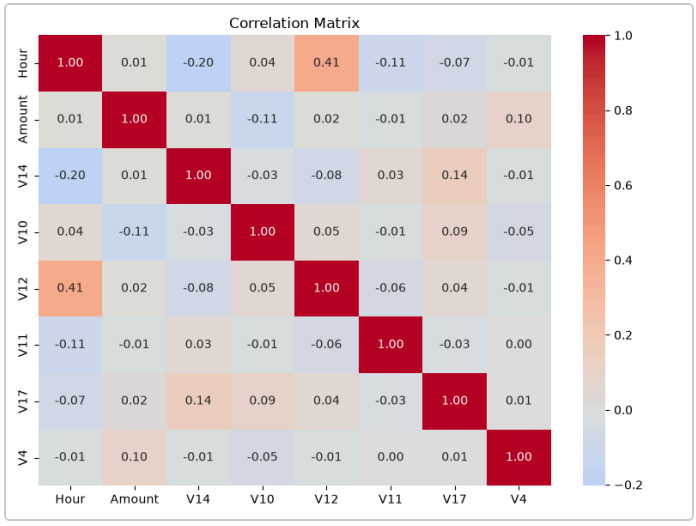


Fig. 4. Correlation matrix of Hour, Amount, and top separating PCA features.

D. Multimodality and Normality Rejection

We applied the D'Agostino-Pearson omnibus test [8] on legitimate transactions for Amount, V1, V4, and V14. Every tested feature yielded p-values $< 10^{-100}$, rejecting unimodal Gaussian behavior. Legitimate consumer spending consists of distinct behavioral personas (e.g., micro-purchases, recurring utilities, commercial corporate cards), necessitating a multi-component mixture model.

V. METHODOLOGY: GMM DENSITY SCORING

A. Mathematical Formulation

A Gaussian Mixture Model models the probability density function of continuous feature vector $x \in \mathbb{R}^D$ as a convex combination of K multivariate Gaussian distributions:

$$p(x \mid \theta) = \sum_{k=1}^{K} \pi_k \, N(x \mid \mu_k, \Sigma_k)$$

subject to mixing proportions $\sum \pi_k = 1$ ($\pi_k \geq 0$), where component density is defined as:

$$N(x \mid \mu_k, \Sigma_k) = (2\pi)^{(-D/2)} \, |\Sigma_k|^{(-1/2)} \exp(-\tfrac{1}{2} (x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k))$$

Model parameters $\theta = \{\pi_k, \mu_k, \Sigma_k\}$ are estimated on unlabelled legitimate training data via the Expectation-Maximization (EM) algorithm [5] with regularisation covariance $\varepsilon = 10^{-5}$ added to diagonals to prevent numerical collapse.

B. Anomaly Scoring Function

Transactions residing in low-density probability regions reflect anomalous behavioral deviations. We compute the anomaly score S(x) as the Negative Log-Likelihood (NLL):

$$S(x) = -\ln p(x \mid \theta) = -\ln \left[\sum_{k=1}^{K} \hat{\pi}_k \, N(x \mid \hat{\mu}_k, \hat{\Sigma}_k) \right]$$

High values of S(x) trigger security triage alerts.

C. Hyperparameter Grid Protocol

The search space evaluated: Feature sets (All 29 vs Selected 12), Components $K \in \{1, 2, ..., 8\}$, and 3 random seeds ($\{0, 1, 2\}$) with n_init = 5 restarts (144 fits total). Models were selected based on validation Average Precision (AP), with Bayesian Information Criterion (BIC) monitored for complexity penalization.

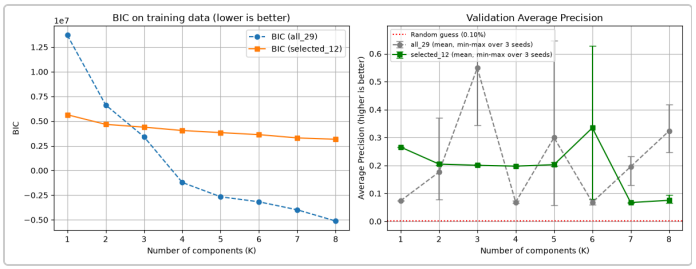


Fig. 5. Hyperparameter search across component count K: (Left) BIC trajectories; (Right) Validation Average Precision across random seeds.

TABLE II: VALIDATION PERFORMANCE ACROSS CONFIGURATIONS

Features	K	AP Mean	AP Min	AP Max	Mean BIC
All 29	1	0.074	0.074	0.074	13,701,380
All 29	2	0.176	0.079	0.371	6,635,476
All 29	**3**	**0.549**	**0.343**	**0.666**	**3,412,903**
All 29	4	0.068	0.065	0.074	-1,218,040
All 29	5	0.300	0.058	0.647	-2,666,021
All 29	6	0.068	0.059	0.078	-3,176,066
All 29	7	0.196	0.128	0.232	-3,987,216
All 29	8	0.323	0.246	0.419	-5,122,966
Sel. 12	1	0.266	0.266	0.266	5,643,466
Sel. 12	2	0.205	0.201	0.213	4,670,746
Sel. 12	3	0.201	0.199	0.205	4,387,461
Sel. 12	4	0.197	0.197	0.198	4,047,104
Sel. 12	5	0.203	0.198	0.212	3,833,669
Sel. 12	6	0.334	0.081	0.628	3,628,444
Sel. 12	7	0.067	0.065	0.069	3,297,546
Sel. 12	8	0.075	0.065	0.095	3,158,924

VI. EXPERIMENTAL RESULTS AND DISCUSSION

A. Validation Findings

As summarized in Table II and Fig. 5:

1. **Failure of BIC for Fraud Ranking:** For All 29 features, BIC turns negative for $K \geq 4$ as Gaussian components collapse onto localized

data pockets, inflating likelihood without improving fraud ranking. Selection must rely on validation AP.

2. **Dimensionality vs. Flexibility Trade-off:** At K=1, Selected 12 features outperform All 29 (AP = 0.266 vs 0.074). However, at K=3, All 29 reaches the study's peak (AP_mean = 0.549, peak seed AP = 0.6661). The 29×29 full covariance matrix captures 406 pairwise relationships that univariate ranking discarded.
3. **EM Solution Variance:** GMM quality varies substantially with initialization seed. For All 29 at K=3, AP ranges from 0.343 to 0.666, demonstrating the necessity of multi-seed evaluations.

Champion Model: All 29 features, K = 3, Full Covariance, Seed 2 (Validation AP = 0.6661).

B. Operational Decision Thresholding

The operational threshold τ was calibrated on validation under Scenario A (False Alarm Budget): $\tau = Percentile_99.5(S_normal) = 94.68$, guaranteeing that ≥99.5% of legitimate cardholders experience uninterrupted service (FPR ≤ 0.5%).

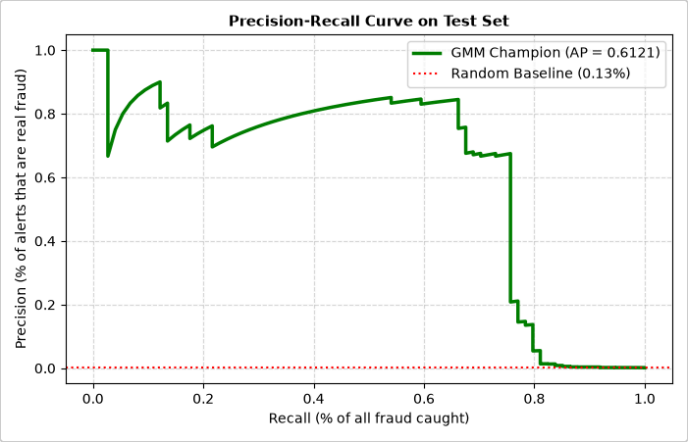


Fig. 6. Precision-Recall Curve on the held-out test set (AP = 0.6121), operating at 470× the random baseline (0.13%).

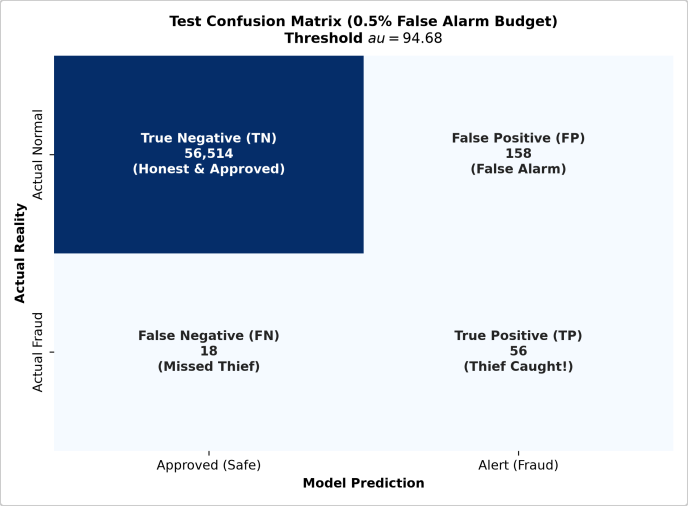


Fig. 7. Test Confusion Matrix evaluated at the frozen validation threshold τ = 94.68 (0.5% False Alarm Budget).

C. Out-of-Time Test Set Evaluation

Applying the frozen champion GMM and threshold τ = 94.68 to the 56,746 untouched test transactions yielded the results in Table III.

TABLE III: TEST PERFORMANCE AND 95% BOOTSTRAP CIS

Metric	Test Value	95% Bootstrap CI
Random Baseline AP	0.0013 (0.13%)	—
Average Precision (AP)	**0.6121**	**[0.485, 0.743]**
PR-AUC (Trapezoidal)	0.6083	—
ROC-AUC	**0.9229**	[0.879, 0.962]
Recall @ τ = 94.68	**75.68% (56/74)**	**[65.79%, 84.72%]**
Realized False Alarm Rate	**0.28%** (158/56,672)	—
Precision @ Top 25 Alerts	**72.00%** (18/25)	—
Precision @ Top 50 Alerts	**84.00% (42/50)**	**[70.00%, 94.00%]**
Precision @ Top 100 Alerts	**56.00%** (56/100)	—

Generalization: Test AP (0.6121) closely tracks validation AP (0.6661) and falls well within the bootstrap interval [0.485, 0.743], confirming strong generalization without overfitting across time boundaries.

Threshold Verification: At τ = 94.68, the model catches 56 of 74 frauds (75.7% Recall) with only 158 false alarms (0.28% FPR), safely satisfying the bank's 0.5% budget (Fig. 7).

Triage Efficiency: In human review queues, the top 50 flagged swipes achieve 84.0% precision (42 confirmed frauds), minimizing investigation waste. Precision moderates to 56.0% at Top 100 as the finite pool of 74 frauds is exhausted.

VII. CONCLUSION & PHASE 2 ROADMAP

This Phase 1 study establishes that unsupervised Gaussian Mixture Models provide a viable frontline triage filter for credit card fraud. By scoring transactions via negative log-likelihood under legitimate data, our pipeline achieved AP = 0.6121 (470× random guess), captured 75.7% of out-of-time fraud at 0.28% FPR, and achieved 84% precision in top-50 review queues.

Phase 2 Transition: In Phase 2 (Deep Learning), we will implement deep autoencoder architectures to benchmark reconstruction error against GMM density scoring, conduct ablation studies, and investigate hybrid autoencoder-GMM representations.

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